Quantum Graphity as a subset of FFGFT

Reduction steps and structural correspondence

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Preamble

Documents 230 and 231 showed that standard quantum mechanics (Hilbert-space formalism) is a *subset* of FFGFT, arising from the full geometry through a reduction chain

$$S^2 \subset \text{cylinder} \subset T^2 \subset T^3 \subset T^4.$$

This document applies the same methodological strategy to another established theory: **Quantum Graphity** (Konopka, Markopoulou, Severini, 2008).

Quantum Graphity is a peer-reviewed, background-independent model in which spacetime and geometry arise as emergent properties of a fundamental *graph*. It is mathematically clearly defined, published in *Phys. Rev. D* 77, 104029 (2008), and thus a concrete object of comparison.

Central thesis of this document: Quantum Graphity *can plausibly be sketched as a subset of FFGFT*, through a five-stage reduction chain from the full FFGFT geometry – analogous to the Bloch sphere, but through different reduction steps. Unlike Doc. 230, however, this translation is *not carried out* but a hypothesis:

the mathematical triangulation construction and the associated convergence proof are open research tasks (see §10.4 below). What is methodologically important already is that FFGFT shares with Quantum Graphity (and LQG, CDT, Wolfram hypergraphs) the basic programme “emergent geometry from discrete microstructure” – without inheriting their limitations.

Notation

Quantum Graphity symbols

Symbol	Meaning
K_N	complete graph on N vertices (all vertices connected)
N	number of vertices (typically $N \gg 1$)
V_a	vertex (node) a
e_{ab}	edge between vertex a and b
$n_{ab} \in \{0, 1\}$	occupation number of edge e_{ab} (off/on), base model
$\|0\rangle, \|1\rangle$	edge states: off/on
$a_{ab}^\dagger, a_{ab}$	creation and annihilation operators on edge e_{ab}
$\{a, a^\dagger\}$	fermionic anticommutator, $= 1$
$N_{ab} = a_{ab}^\dagger a_{ab}$	number operator (adjacency matrix element)
$\|j, m\rangle$	extended edge labels ($j = 0, 1$, $m \in \{-j, \dots, j\}$)
$M, M^\pm$	angular momentum operators on $\|j, m\rangle$ labels
$\mathcal{H}_{\text{edge}}$	Hilbert space of a single edge
$\mathcal{H}_{\text{total}}$	tensor product Hilbert space of all edges
$H_V, H_B, H_C, H_D, H_\pm$	Hamiltonian components (valence, cycles, C, D, loop)
$g_V, g_B, g_C, g_D, g_\pm$	coupling constants of Hamiltonian terms
v_0	preferred vertex valence (typically $v_0 = 3$ for hexagonal)
S_N	permutation group on N elements

FFGFT symbols (recalled from Doc. 230/231)

Symbol	Meaning
T^n	n-torus, n compact loops
T^4	full FFGFT geometry incl. time winding
(n_x, n_y, n_z, k_t)	FFGFT winding quantum numbers
$\varphi_{n,l,j}$	FFGFT mode function on T^4
$\xi_0 = 4/30000$	FFGFT geometry parameter
$L_0 = \xi_0 \cdot L_P$	FFGFT minimum length scale (sub-Planck)
L_P	Planck length
$U(1)$	electromagnetic gauge group

Other symbols

Symbol	Meaning
$\subset$	subset of, contained in
$\to$	reduction or transition
$\beta = 1/k_B T$	inverse temperature
T_c	critical temperature (phase transition)

1 Quantum Graphity in one page

Quantum Graphity (Konopka, Markopoulou, Severini, *Phys. Rev. D* 77, 104029 (2008), arXiv:0801.0861) is a background-independent model in which space and geometry emerge from a dynamical *graph*. The basic assumptions are:

- **Complete graph K_N:** on $N \gg 1$ vertices, *all* vertex pairs are connected by edges. K_N has $\binom{N}{2}$ edges.
- **Edge Hilbert space (base model):** each edge e_{ab} carries a Hilbert space $\mathcal{H}_{\text{edge}} = \text{span}\{|0\rangle, |1\rangle\}$ – a

fermionic oscillator (or equivalently a qubit). $|0\rangle$ is "off", $|1\rangle$ is "on".

- **Tensor-product Hilbert space:** the total Hilbert space is $\mathcal{H}_{\text{total}} = \bigotimes_{a<b} \mathcal{H}_{\text{edge}}$, i.e. tensor product over all $\binom{N}{2}$ edges.
- **Adjacency operators:** $N_{ab} = a^{\dagger}_{ab} a_{ab}$ is the quantum-mechanical adjacency matrix; the active edges form a subgraph of K_N.
- **Hamiltonian:** several permutation-invariant terms (H_V valence, H_B cycles, etc.) drive a *phase transition*.
- **High-energy phase:** all edges on, full permutation symmetry S_N, no notion of locality.
- **Low-temperature phase:** permutation symmetry breaks to translation symmetry. A *regular subgraph* emerges – in the standard case a *hexagonal* (3-regular) configuration.
- **Extension with "matter labels" $|j, m\rangle$:** in §IV of the paper the base model is extended: each edge can carry $j = 0, 1$ and $m \in \{-j, \dots, +j\}$. This extension allows *emergent matter* and *emergent $U(1)$ gauge theory* through string-net condensation (Levin-Wen mechanism).

What Quantum Graphity does not do: it does not deliver Standard-Model constants, fermion masses, or the fine-structure constant. It delivers the skeleton "how geometry emerges from discrete microstructure", plus an emergent $U(1)$ gauge theory, without specifically explaining the SM matter structure.

2 Methodological parallel to Hilbert-space reduction

In Doc. 230 we showed how the standard Bloch sphere of quantum mechanics arises from FFGFT through a geometric reduction ($T^4 \to T^3 \to T^2 \to$ cylinder $\to S^2$). At each step a specific structural layer was lost – fractal corrections, winding quantum numbers, bundle structure, time winding.

Here we exhibit the analogous reduction for Quantum Graphity:

$$\boxed{\text{Quantum Graphity} \subset \text{FFGFT}} \tag{1}$$

through a different – but structurally comparable – reduction chain. The reduction steps are:

1. **Discretisation:** continuous $T^4 \to$ discrete graph with N vertices.
2. **Topology forgetting:** T^4 topology $\to$ abstract complete graph K_N.
3. **Quantum-number generalisation:** FFGFT windings $(n_x, n_y, n_z, k_t) \to$ general edge occupation / spin labels.
4. **Parameter generalisation:** $\xi_0 = 4/30000 \to$ general coupling parameters.
5. **Matter dropping:** 12 fermion masses + $\alpha \to$ no counterpart.

These five reductions together yield the Quantum-Graphity skeleton. They do *not* correspond to the Bloch-sphere reduction of Doc. 230 – they are a *different* projection of the same FFGFT full geometry onto a different abstract description.

3 Reduction 1: continuum to discrete

3.1 FFGFT: continuous windings on T^4

In FFGFT the full geometry lives on a 4-torus $T^4 = T^3 \times S^1$. Mode functions $\varphi_{n,l,j}(x_0, x_1, x_2, t)$ are *smooth functions* on this compact manifold. The mode quantum numbers (n_x, n_y, n_z, k_t) are discrete – but each mode is a continuous function, and the underlying geometry is differential-geometrically smooth.

3.2 Quantum Graphity: discrete graph

In Quantum Graphity there is no smooth manifold. There is only the discrete complete graph K_N with N vertices and $\binom{N}{2}$ possible edges. Each edge is either active or inactive; active edges may carry labels.

3.3 Reduction

Discretisation *would* be sketched formally as triangulation of T^4: subdividing T^4 into N small cells (e.g. tetrahedra or small cubes) yields a graph whose vertices correspond to cells and whose edges encode neighbourhood relations. Heuristically, as the mesh becomes fine ($N \to \infty$, cell size $\to 0$), this graph should converge to the continuous manifold.

Important caveat: this construction is a *sketch*, not a carried-out proof. A rigorous triangulation translation FFGFT $\to$ Quantum Graphity has not been worked out in the FFGFT documents – it is an open research task (see §10.4 below). The methodological plausibility that such a construction is possible rests on the established tradition of simplicial approximation in algebraic topology. A concrete convergence theorem for the FFGFT mode functions $\varphi_{n,l,j}$ under a specific T^4 triangulation, however, is not proved.

What is lost: differential geometry. The FFGFT equation of motion $\mathcal{D}_{T^4}\varphi = E\varphi$ (Dirac-like operator on T^4) has no direct counterpart in the discrete picture – one obtains instead a Hamiltonian update rule on the edges.

What remains: the topological information *if* one remembers that the graph came from a 4-torus triangulation. In Quantum Graphity, however, this memory is *deliberately forgotten* (see Reduction 2).

4 Reduction 2: forgetting topology

4.1 FFGFT: T^4 topology

The 4-torus has a very specific topology: four compact loops, each with winding number $\in \mathbb{Z}$. This topology is *part of the physical statement* of FFGFT – the winding quantum numbers are not arbitrary but correspond to specific physical quantities (masses, charges, spin).

4.2 Quantum Graphity: complete graph K_N

In Quantum Graphity one starts with the *complete* graph K_N – thus with *maximal* connectivity, without any topology. This is the high-temperature state. Only the phase transition at low temperature lets a low-dimensional topology emerge.

From the FFGFT perspective: the complete graph K_N corresponds to "forgetting" the T^4 topology. If one no longer remembers that the N points were originally organised in a 4-torus geometry, every point can be connected to every other – yielding K_N.

4.3 Reduction

Heuristically the step can be described as: starting from a (sketched) triangulation of T^4 from Reduction 1, one would have a *local* graph G_{T^4} in which each vertex has only nearby neighbours. Stripping this locality information by declaring all vertex pairs as potentially connectable would yield K_N.

This step too remains a sketch. The concrete construction of a map $G_{T^4} \to K_N$ and the proof that under this map the Quantum-Graphity Hamiltonian components (H_V, H_B, etc.) reproduce the correct FFGFT low-energy physics are not worked out.

What is lost: locality. In FFGFT, locality is given (by the T^4 geometry). In Quantum Graphity, locality is an *emergent phenomenon* of the low-temperature phase.

Methodological point: Quantum Graphity treats locality as emergent. FFGFT treats it as given. This is not a contradiction but a difference of description level – in FFGFT the underlying geometry is already chosen; in Quantum Graphity the choice of geometry is left to the system.

5 Reduction 3: windings to spin labels

5.1 FFGFT: specific winding quantum numbers

In FFGFT the winding quantum numbers $(n_x, n_y, n_z, k_t) \in \mathbb{Z}^4$ are *physically interpreted*:

- (n_x, n_y, n_z) encode spatial mode position and angular momentum (Doc. 210),
- k_t carries the mass energy of the mode ($mc^2 = \hbar f(k_t)$),
- specific tuples yield specific Standard-Model particles.

5.2 Quantum Graphity: fermionic occupation and (j, m) labels

In Quantum Graphity the edges in the *base model* carry only a fermionic occupation number $n_{ab} \in \{0, 1\}$ – off or on. In the *extended version* (§IV of the paper) the "on" states acquire additional labels $|j, m\rangle$ with $j = 0, 1$ and $m \in \{-j, \dots, +j\}$. The labels are general *spin-network-like* and belong to the LQG tradition. The operators $M, M^{\pm}$ obey the usual angular-momentum algebra

$$[M^+, M^-] = M, \qquad [M, M^{\pm}] = \pm M^{\pm}.$$

5.3 Reduction

The reduction FFGFT $\to$ Quantum Graphity in this layer is *forgetting the specific physical meaning* of the winding quantum numbers. The specific tuple (n_x, n_y, n_z, k_t) that for example describes an electron is replaced by:

- In the base model: only "on"/"off" per edge – no quantum-number information at all.
- In the extended version: (j, m) labels carrying general spin-1 information, without specifying which number carries which physical quantity.

What is lost: the connection to the Standard Model. In FFGFT the winding structure directly yields the mass formula $m_i = r_i \xi^{p_i} v$ and thus all 12 fermion masses. In Quantum Graphity there is no mechanism that links specific (j, m) values to specific masses.

6 Reduction 4: geometry parameter to phase parameters

6.1 FFGFT: $\xi_0 = 4/30000$ is geometrically fixed

In FFGFT the geometry constant $\xi_0 = 4/30000$ is not a free parameter, but follows from Higgs matching $\xi_0 = \lambda_h^2 v^2 / (16\pi^3 m_h^2)$ (Doc. 006). It is geometrically and physically *uniquely fixed*.

6.2 Quantum Graphity: multiple coupling parameters

In Quantum Graphity there is a whole set of coupling parameters: $g_V, g_B, g_C, g_D, g_\pm$ (Hamiltonian components) plus v_0 (preferred vertex valence) and p, r. Their values determine where the phase transition occurs, which subgraph emerges in the low-temperature phase (typically $v_0 = 3$ yields a *hexagonal* configuration), and how strong the string-net condensation is. None of these parameters is fixed externally – they are *free*.

6.3 Reduction

The transition FFGFT $\to$ Quantum Graphity in this layer is the *multiplication and generalisation of parameters*. The single geometrically fixed quantity $\xi_0 = 4/30000$ is replaced by a whole set of free couplings.

What is lost: predictive power. The values of all SM constants in FFGFT follow from the single parameter ξ_0. If one replaces ξ_0 by several free couplings, these predictions are lost – one can determine the phase diagram, but not the values of physical constants.

Methodological point: Quantum Graphity shows that a family of parameters suffices to generate a rich emergent geometry. FFGFT shows additionally that all these parameters can be fixed by a single geometric quantity (ξ_0) – and then also yield all SM constants.

7 Reduction 5: dropping the matter sector

7.1 FFGFT: full matter sector

In FFGFT, all 12 fermion masses, the fine-structure constant $\alpha = \xi \cdot E_0^2$, the Higgs self-coupling, and the spin-orbit constants of atomic physics follow from the winding quantum numbers (Doc. 006, 174, 210). The matter sector is not introduced separately, but *integral part* of the geometry.

7.2 Quantum Graphity: no matter sector (in the base model)

In the base model of Quantum Graphity there is no matter specification – only the binary edge states $|0\rangle, |1\rangle$ and the emergent lattice. In the *extended version* with (j, m) labels a $U(1)$ *gauge field* emerges (through the string-net condensation of Levin-Wen) – this is analogous to a photon field. But there are no fermions, no quarks, no specific masses – the string-net condensation does generate point-like excitations with mass $\propto g_C$, but these masses are free parameters, not determined by a geometric source.

7.3 Reduction

The final reduction step is the complete *discarding* of the matter-determining FFGFT sector. FFGFT contains matter as winding modes on T^4 with *fixed mass relations*. Quantum Graphity in the extended version contains emergent $U(1)$ gauge fields and weakly bound excitations, but without a geometric mass formula.

What is lost: everything FFGFT delivers for the explanation of SM masses and couplings.

What remains: the emergent gauge theory. The fact that $U(1)$ emerges in both theories from microstructure is a real structural agreement – it is the only row in the translation table that truly *exactly* corresponds.

8 Translation table

FFGFT structure	Quantum Graphity counterpart	Reduction
4-torus T^4	complete graph K_N	discretisation + topology forgetting
windings (n_x, n_y, n_z, k_t)	occupation n_{ab} + ext. (j, m)	specific meaning lost
mode functions $\varphi_{n,l,j}$	edge states $\|0\rangle, \|1\rangle$	smooth $\to$ discrete
$\xi_0 = 4/30000$ single constant	$g_V, g_B, g_C, g_D, g_\pm$ and v_0 free	predictive power lost
low-energy smoothing to T^4 geometry	low-temperature: hexagonal 3-regular	same principle
emergent $U(1)$ from EM winding	emergent $U(1)$ via Levin-Wen (ext. only)	true correspondence
12 fermion masses	—	no counterpart
$\alpha = \xi E_0^2$	—	no counterpart
spin-orbit constants	—	no counterpart

Status of the correspondences:

- **One real correspondence:** emergent $U(1)$ in both theories from microstructure (in QG only in the extended version with (j, m) labels).

- **Structural parallels:** low-energy smoothing to ordered geometry is the same basic principle in both theories.
- **Reduction losses:** all SM-specific quantities (masses, α, spin-orbit) have no Quantum-Graphity counterpart – they are lost in the reduction process.

9 What is lost in reduction – and what remains

9.1 What is lost

The five reduction steps lead to a progressive loss:

- differential geometry (Reduction 1),
- locality as a given structure (Reduction 2),
- physical meaning of the winding quantum numbers (Reduction 3),
- geometry-parameter specificity (Reduction 4),
- entire matter sector (Reduction 5).

9.2 What remains

Remarkably, *the essential geometry-emergence scheme* is preserved:

- In FFGFT: the full geometry is not directly visible but emerges from the mode configuration in the low-energy limit.
- In Quantum Graphity: geometry is not laid down in advance but emerges from the phase transition in the low-temperature phase.

Both theories share the same methodological position: geometry is not given but emergent. FFGFT specifies *which* geometry (T^4) and *why* (geometric Higgs matching, ξ_0 quantisation). Quantum Graphity shows that this idea of an emergent geometry from discrete microstructure is *not exotic*, but is pursued in established peer-reviewed quantum-gravity research.

10 Methodological position

10.1 What this reduction shows

The five reduction steps show that Quantum Graphity is a *subset of FFGFT* – analogous to the Bloch sphere, but through different reduction steps. Concretely:

- Bloch sphere $\subset$ FFGFT through *geometric reduction* (pole contraction, loop cutting).
- Quantum Graphity $\subset$ FFGFT through *structural reduction* (discretisation, topology forgetting, matter dropping).

10.2 Why this matters

This strategy – a *subset relation* instead of a full bijection – is methodologically important:

- It is **honest:** no claim of an exact translation where only a partial one is possible.
- It **positions FFGFT as the more comprehensive framework:** Quantum Graphity is a special case, not a competitor.
- It **connects FFGFT to established research:** FFGFT shares the methodological core programme with Konopka, Markopoulou, Severini – and thus also with Loop Quantum Gravity, Causal Dynamical Triangulation, Wolfram hypergraphs.
- It **shows FFGFT's added value:** while Quantum Graphity delivers only emergent geometry, FFGFT delivers additionally the Standard-Model constants.

10.3 Connection to Doc. 230 and 231

- **Doc. 230** showed: Hilbert-space QM $\subset$ FFGFT (through geometric reduction to the Bloch sphere). This reduction is *carried out* – the translation bridge $\alpha = \sqrt{(1+z)/2}\, e^{i\theta/2}$ is a concrete bijection.

- **Doc. 231** showed: Hilbert-space QM can be *extended* to full FFGFT (through four mathematical structures: $SU_q(2)$, winding quantum numbers, spinor bundles, Kaluza-Klein). These extensions are *established* – $SU_q(2)$ exists since 1985, spinor bundles since the 1950s, Kaluza-Klein since 1921.
- **Doc. 232 (this document)** sketches: Quantum Graphity $\subset$ FFGFT (through structural reduction). This reduction is *not yet carried out* – it is plausibly sketched, but not rigorously constructed (see §10.4).

Together the three documents yield the picture: **FFGFT demonstrably contains Hilbert-space QM as a subset (Doc. 230/231) and plausibly contains the discrete-geometry programs as special cases, provided the triangulation reduction can actually be carried out (Doc. 232).** It is not a competitor to these theories, but the relationship to Hilbert-space QM and to Quantum Graphity is *not symmetric* – see §10.4 and §11.

10.4 Status of the reduction: hypothesis, not proof

Important caveat. At several points in the FFGFT documents – including this one – there is talk of "triangulations", "tetrahedron geometry", and "translatability into triangle or network descriptions". This language is useful for locating FFGFT methodologically within the broader quantum-gravity tradition (LQG, CDT, Wolfram, Quantum Graphity), but it must not be taken as a proved mathematical translation theorem.

What actually holds:

- *Methodological kinship:* FFGFT shares with Quantum Graphity (and LQG, CDT) the basic programme "emergent geometry from discrete microstructure". This is a conceptual statement, not a mathematical theorem.

- *Plausible reduction chain:* the five reduction steps in §3-7 of this document are a plausible sketch of how a translation FFGFT $\rightarrow$ Quantum Graphity *could look*. It is consistent with standard differential topology (simplicial approximation) and with the structures of the Quantum-Graphity Hilbert space (fermionic tensor product over edges).
- *Internal FFGFT tetrahedral geometry:* The tetrahedron factor $4/3$ in $\xi_0 = 4/30000$ is an *internal* FFGFT statement (Doc. 180, 202), not a claim about translatability to LQG spin-foam models.

What remains open:

- A *rigorous convergence proof* that a specific T^4 triangulation with $N \rightarrow \infty$ approximates the FFGFT mode functions $\varphi_{n,l,j}$.
- A *Hamiltonian translation theorem* relating the FFGFT operator $\mathcal{D}_{T^4}$ to the Quantum-Graphity Hamiltonian chain $H_V + H_B + \cdots$.
- A *recovery theorem* showing that the low-temperature phase of a Quantum-Graphity model with appropriately chosen couplings yields an effective T^4.

Methodological consequence: The statement "Quantum Graphity $\subset$ FFGFT" is a *plausible programme item*, not a proved structural statement – in contrast to the Hilbert-space translation in Doc. 230, which exists as a concrete bijection. This asymmetry between Docs 230 and 232 must be kept in mind when both documents are used. Anyone presenting FFGFT's relationship to LQG, CDT, Wolfram or Quantum-Graphity programmes should formulate it as *methodological kinship with shared basic assumptions* – not as a formally proved reduction.

11 Comparison of explanatory power: Hilbert-space QM vs. Quantum Graphity

The reduction FFGFT $\to$ Quantum Graphity differs *fundamentally* from the reduction FFGFT $\to$ Hilbert-space QM (Doc. 230) – not only in the reduction steps, but in the **explanatory power** of the respective target theory. This asymmetry is methodologically important and is made explicit here.

11.1 Hilbert-space QM: substantial content

Standard Hilbert-space quantum mechanics is a *mathematically mature, complete* theory:

- It describes matter completely (wave functions, spinors, many-particle states).
- It yields precise predictions (atomic spectra, scattering amplitudes, quantum gate operations).
- It is overwhelmingly experimentally confirmed.
- It accommodates all three gauge groups $SU(3) \times SU(2) \times U(1)$ in tensor-product form.
- It has a century of mathematical development behind it.

The translatability FFGFT $\leftrightarrow$ Hilbert-space QM (Doc. 230) is therefore a *bridge between two mature theories*, with small, quantifiable loss (K_{frak} correction, ΔCHSH $\approx 10^{-5}$).

11.2 Quantum Graphity: programmatic content

Quantum Graphity is a *programme*, not a complete tool:

- Matter is not described in the base model; in the extended version only as (j, m) labels without mass specificity.
- Predictions are *qualitative* (phase transition, emergent locality), not quantitative.

- There are no direct experimental confirmations – Quantum Graphity operates at the Planck scale.
- Only one gauge group ($U(1)$) emerges, and only in the extended version with Levin-Wen string-net condensation.
- It is 17 years old (first paper 2008), with a small research community.

11.3 Quantitative comparison

Aspect	**Hilbert-space QM**	**Quantum Graphity**
Matter sector	fully described	not present (base model)
Masses	external Yukawa parameters	not present
Gauge theories	$SU(3)\times SU(2)\times U(1)$	only $U(1)$ (ext. version)
Spin algebra	complete	added if needed
Predictions	precise quantitative	qualitative
Experimental confirmation	overwhelming	none direct
Mathematical maturity	one century	17 years
Reduction loss vs. FFGFT	small, quantifiable	massive

11.4 What the asymmetry means for FFGFT's position

The two reductions show FFGFT's relationship to two very different theory classes:

- **Towards Hilbert-space QM:** FFGFT has to respect a *large, established* theory – and does so, with full translatability and small geometric corrections. This is FFGFT's proof of *consistency* with known physics.
- **Towards Quantum Graphity:** FFGFT contains a *small, programmatic* model as a special case. This is FFGFT's proof of *methodological kinship* with other quantum-gravity programmes – without FFGFT having to inherit their limitations.

The point: Quantum Graphity is *not wrong*, but *little*. FFGFT shares with it the basic methodological programme "emergent geometry from discrete microstructure", but goes far beyond: it specifies the geometry (T^4), the single parameter (ξ_0), and from these yields all 12 fermion masses, the fine-structure constant, the gauge-group structure, and the spin-orbit constants – all parameter-free.

11.5 Strategic consequence

- **To Hilbert-space practitioners (mathematicians, QM theorists):** FFGFT respects full standard QM and extends it consistently (Doc. 230, 231). The translation tables are interoperable.
- **To quantum-gravity researchers (LQG, Wolfram, Quantum Graphity):** FFGFT shares the basic programme "emergent geometry from microstructure" but goes further and yields SM constants parameter-free. Quantum Graphity is a special case in which almost all FFGFT statements are dropped.

FFGFT's strength arises precisely from this asymmetry: it is *large enough* to contain Hilbert-space QM, and *specific enough* to extend Quantum Graphity.

12 Summary

Quantum Graphity is plausibly representable as a subset of FFGFT – in the sense of a sketched, not carried-out reduction.

The reduction chain FFGFT $\to$ Quantum Graphity consists of five steps:

1. Discretisation of the T^4 continuum (sketched as a triangulation – not yet rigorously constructed).
2. Forgetting the T^4 topology in favour of the complete graph K_N.
3. Generalisation of the winding quantum numbers (n_x, n_y, n_z, k_t) to general edge states (n_{ab} in the base model, (j, m) in the extended version).
4. Generalisation of the geometry parameter $\xi_0 = 4/30000$ to a set of free couplings $g_V, g_B, g_C, g_D, g_\pm, v_0, \ldots$.
5. Dropping the matter sector (masses, α, spin-orbit).

What remains after reduction: the *geometry-emergence scheme* and the *emergent* $U(1)$ *gauge theory*. What is lost: the SM-specific predictions.

Methodological position: FFGFT shares with Quantum Graphity (and LQG, CDT, Wolfram hypergraphs) the basic programme "emergent geometry from discrete microstructure". FFGFT goes further: it specifies the geometry (T^4), the parameter (ξ_0), and from that yields all SM constants parameter-free.

Status of the reduction: In contrast to the Hilbert-space translation in Doc. 230 (concrete bijection with $\alpha = \sqrt{(1+z)/2}\, e^{i\theta/2}$), the reduction in this document is a *hypothesis* – plausibly supported by standard differential topology, but without a carried-out convergence proof. The statement "Quantum Graphity $\subset$ FFGFT" is therefore to be understood as a *methodological programme item*, not as a formally proved structural statement.

Quantum Graphity is therefore *not a contradiction* of FFGFT but a *partial FFGFT skeleton* – the emergent-geometry component without the matter specificity. The existence of Quantum Graphity as a peer-reviewed research programme demonstrates that FFGFT's methodological assumptions are *not exotic*.